CBM CONDITION	80
	DEFECT

DISTRIBUTION SUBSTATION (PCE) CBA TEST SHEET

STATE / AREA:	JOHOR / JOHOR BAHRU	**DATE**	14/07/26			**FUNCTIONAL LOCATION:**	
PE NAME:	SK KG MAJU JAYA	**TIME IN / OUT**	1940	/	2030	JJBU/PCE/J03951	
V.I REF. NO	80	**TEV BACK GROUND / HUMIDITY**	5 dB	/	63 %	**BACKGROUND TEMP : 32.0 °C**	

SWITCHGEAR

DETAIL								TEMPERATURE							ULTRASOUND			TEV			REMARK
NO.	FDR. NO	FEEDER NAME	STATUS (TRIP / CLOSE)	LOAD (A)	CABLE TYPE/SIZE	HEATER (A)	SERIAL NO (ONLY BREAKER & OLU)	COMP	Tmin (°C)	Tmax (°C)	ΔT	Avg (°C)	FILE	REMARKS	dB	FILE	TYPE CHAR.	dB	PULSE/ CYCLE	TYPE CHAR.	
1	JJB96243	ULAM RAJA	CLOSE		XLPE			CABLE	32.6	32.8	0.2	32.7	227		0			5	0		
								BREAKER													
								TOP PANEL													
								PT													
2	JJB96244	TX 1000 KVA	CLOSE		XLPE			CABLE	32.4	32.6	0.2	32.5	228		0			5	0		
								BREAKER													
								TOP PANEL													
								PT													
3	JJB96245	SETIA B/PARK 2 NO 15	CLOSE		XLPE			CABLE	32.6	32.6	0.0	32.6	229		0			5	0		
								BREAKER													
								TOP PANEL													
								PT													
4								CABLE													
								BREAKER													
								TOP PANEL													
								PT													
OVERVIEW								CABLE				33.2	222								
								BREAKER													
								TOP PANEL				32.2	226								
								PT													

LOCAL TRANSFORMER

DETAILS				THERMOGRAHY					ULTRASOUND		DETAILS				THERMOGRAHY						ULTRASOUND		
TX ID	SERIAL NO	CABLE TYPE/SIZE	REMARK	COMP.	Tmin(°C)	Tmax(°C)	ΔT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE/SIZE	COMP.	Tmin(°C)	Tmax(°C)	ΔT	Avg (°C)	FILE	dB	FILE	TYPE CHAR.	REMARKS
TX1	EWT/91670	XLPE		HT CABLE	33.3	33.5	0.2	33.4	233	0		TX3			HT CABLE									
				HT BUSHING	38.3	38.6	0.3	38.5	232	0					HT BUSHING									
		XLPE		LV CABLE	36.0	36.3	0.3	36.2	235						LV CABLE									
				LV BUSHING	36.0	36.2	0.2	36.1	234						LV BUSHING									
				BODY	39.8	44.5	4.7	42.2	230						BODY									
TX2				HT CABLE								TX4			HT CABLE									
				HT BUSHING											HT BUSHING									
				LV CABLE											LV CABLE									
				LV BUSHING											LV BUSHING									
				BODY											BODY									

LV DISTRIBUTION BOARD/FEEDER PILLAR/OTHERS

1	LVDB / FP 1	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10							
	CABLE TYPE / SIZE	XLPE	XLPE		XLPE	XLPE		XLPE	XLPE	XLPE	XLPE										
2	LVDB / FP 2	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10							
	CABLE TYPE / SIZE																				

NO	LVDB / FP	DEFECTIVE FEEDER NAME	Tmin(°C)	Tmax(°C)	ΔT	FILE	Avg (°C)	REMARK
	INC 1							
FPTX1	INC 2	RED PHASE 67	51.9	66.7	14.8	237	59.3	INCOMING LINK CONNECTION
	F1							
	F2							
	F3							
	F4	RED PHASE 108	45.0	106.2	61.2	238	75.6	OUTGOING FUSE CONNECTION
	F5							
	F6	RED PHASE 149	82.6	149.2	66.6	239	115.9	OUTGOING FUSE CONNECTION
	F7	RED PHASE 220	51.6	219.2	167.6	240	135.4	OUTGOING FUSE CONNECTION
	F8							
	Battery Bank					241		BC1: SUNPOWER M:SPS 30-10L SN: 11193-9636
	Black Box 1							
	Black Box 2							
	Lightning Arrester							
	Dropout Fuse							
	Auto Recloser							
	LBS							

FP TX1	J-SLOTTED
Manufacturer :	TOPRANK
Serial No. :	652
Rating :	1600A
LVDB TX2	
Manufacturer :	
Serial No. :	
Rating :	

Tested by (Name & Signature) :

EET ENGINEERING SERVICES SDN BHD
(534556-W)
No. 38-2, 2nd Floor, Jalan SS 19/1D
47500 Subang Jaya, Selangor.
Tel: 603-5636 6201 Fax: 603-5636 7205

TNB Supervisor/Engineer (Name & Signature) :

NOR AZMI BIN ABDUL RAHMAN
(N/P 10013600)
Pen. Juruteknik T/Kanan
Diagnostik (Substation - South)
Distribution Network Department, TNB

Ver. : 1-2019

vi fe miss, light miss, n plock mg

TYPE PCE 80

VISUAL INSPECTION CHECKLIST FOR 11kV SUBSTATION

TENAGA NASIONAL BERHAD (200866-W)

DATE:	14/07/26											
P/E Name (ERMS):	SK KG MAJU JAYA						F/L NO (ERMS):	JJBU/PCE/J03951				
P/E Name (SITE):	SEK. KEB. KG. MAJU JAYA						F/L NO (SITE):	JJBU/PCE/J03951				
GPS Coordinate:	1.579787, 103.718929											
PCE Type:	INDOOR			OUTDOOR	☑	COMPACT		ATTACH BUILDING		PAT		OTHER	
PCE Area	NORMAL	☑		INDUSTRIAL		COASTAL						
Switcgear Type 1:	RMU OIL			RMU SF6		MRMU	☑	VCB		OCB		OTHER	
Manufacturer/Model:	EPE			CFC 2S+1F									
Switchgear Manufactured Date:	2024			Switchgear Rating :			12kV			RMU SF6 / OIL S/N :		23050047400	
Switcgear Type 2:	RMU OIL			RMU SF6		MRMU		VCB		OCB			
Manufacturer/Modal:													
Switchgear Manufactured Date:				Switchgear Rating :						RMU SF6 / OIL S/N :			
No. of Transformer:	1												
Tx 1:	Type	H/S	KVA	1000kVA	Construction Yr:	2024	Manufacturer:	EWT	S/N :	EWT/91670			
Tx 2:	Type		KVA		Construction Yr:		Manufacturer:		S/N :				
Tx 3:	Type		KVA		Construction Yr:		Manufacturer:		S/N :				
Tx 4:	Type		KVA		Construction Yr:		Manufacturer:		S/N :				

KEJANGGALAN

1. Signboard:	Good	☑	Not Good		Remarks	
2. Earth connection:	Good	☑	Not Good		Remarks	
3. EFI Functioning:	Good	☑	Not Good		Remarks	
4. Heater Functioning:	Good		Not Good		Remarks	N/A
5. Labelling @Switchgear:	Good	☑	Not Good		Remarks	
6. Labelling @ LVDB:	Good	☑	Not Good		Remarks	
7. Gas Pressure:	Good	☑	Not Good		Remarks	
8. Tx Guard:	YES	☑	NO		Remarks	
9. Tx Oil Level (conservator tank):	Good		Not Good		Remarks	N/A
10. Tx Oil Leak:	YES		NO	☑	Remarks	
11. Tx Bushing:	Good	☑	Not Good		Remarks	
12. Top Demand Indicator (@LVDB):	Good	☑	Not Good		Remarks	
13. Link/fuse Holder:	Good	☑	Not Good		Remarks	
14. LV Termination:	Good	☑	Not Good		Remarks	
15. Rubber Mat:	YES	☑	NO		Remarks	
16. Trenching:	Good	☑	Not Good		Remarks	
17. Battery Voltage:	Good	☑	Not Good		Remarks	
18. Lighting:	Good		Not Good	☑	Remarks	
19. Roof Condition:	Good	☑	Not Good		Remarks	
20. Fire Fighting	Valid		Expired		Remarks	Expiry Date: - No. of missing portable : 1
21. Fence/door condition:	Good	☑	Not Good		Remarks	
22. Cleanliness/Bushes:	YES		NO	☑	Remarks	
23. Vandalisme	YES		NO	☑	Remarks	
24. Other Findings:						

Tested by (Name & Signature) :

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Date :

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Distribution Network Department, TNB

Date :

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